Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain how field-line drawings represent a field and what they leave out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetic field” and “pole” to explain your answer. Sentence starters are optional.

magnetic field: a field that influences magnets and moving charges

pole: a north or south region of a magnet

superposition: adding displacements or fields at the same place and time

01 Warm-up: bring back an earlier idea

A **5 Ω** resistor is in series with two parallel **6 Ω** resistors across **16 V**. **Find parallel equivalent**, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$.
Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run *north* to *south*.

Worked reasoning: Outside a bar magnet, field arrows run from the *north* pole toward the *south* pole. A compass's *north*-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Sentence starter: Both ... . They differ in ... because ... .

N	S

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Label N** and S on the bar magnet.
2. **Sketch external field arrows** and the direction a nearby compass would point.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

N	S

Marzano check - circle: ① ② ③

04 Apply the physics

A compass between two identical magnets has no stable preferred direction at one location.
1. **Draw two possible opposing field** contributions there.
2. **Explain how superposition could account** for the observation and why it does not prove every field is zero.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

Magnetic fields

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student draws field lines that cross at one point. **Explain why this** is inconsistent with a single net field direction at that point.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

In separate cases, an ideal blackbody temperature becomes **4** times as large and a sound frequency becomes **6** times as large at fixed sound speed. **Compare the two wavelength changes**.
Keep the two relationships separate and **label each ratio**.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: Both … . They differ in … because … .

Case	Quantity changed	Wavelength ratio
Thermal spectrum	Temperature	____
Sound at fixed speed	Frequency	____

Marzano check - circle: 1 2 3

07 Exit check

If a bar magnet is cut in half, do you obtain an isolated *north* pole and an isolated *south* pole? **Explain the usual result**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two magnets face N toward N. **Sketch the field qualitatively** and **predict the force**. Then reverse one magnet and **explain both changes**.

Sentence starter: I predict … . I used … to make this prediction because … .

Label poles; add force or field arrows.



Marzano check - circle: 1 2 3